

# Algorithm Design

- ❑ *What is Algorithm ?*
- ❑ *What is Algorithm Design ?*
- ❑ *Importance of Algorithm Design*
- ❑ *How to Design an Algorithm ?*
- ❑ *Advantages of designing an Algorithm*



# What is Algorithm ?

An algorithm (pronounced AL-go-rith-um) is a procedure or formula for solving a problem.



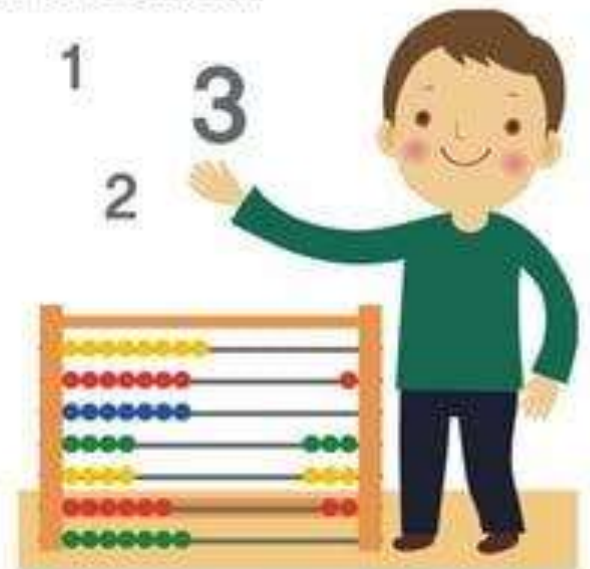
# What is Algorithm Design ?

*It is a specific method to create a mathematical process in solving problems .*



# Importance of Algorithm Design

*It is important because it allows for analysis on different ways to compute things and ultimately come up with the best way to solve a particular problems .*





# How to Design an Algorithm ?

*For making a design of algorithm we need a technique of :*

- *Dynamic programming*
- *Graph algorithms*
- *Divide and conquer*
- *Back tracking*
- *Greedy Algorithm*
- *Flow chart*

# Dynamic Programming

*The dynamic programming is a paradigm of algorithm design in which an optimization problem is solved by a combination of caching subproblem solutions and appealing to the "principle of optimality."*

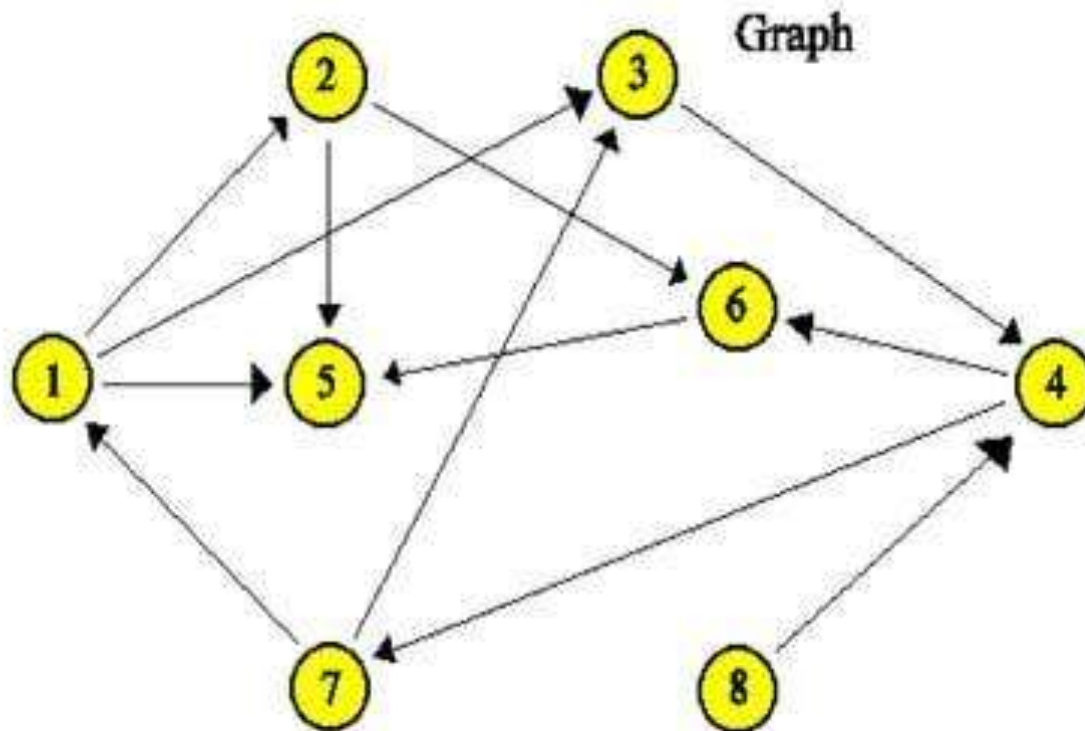


# Basic Elements of Dynamic Programming

1. ***Substructure***
2. ***Table-Structure***
3. ***Bottom-up Computation***

# Graph Algorithm

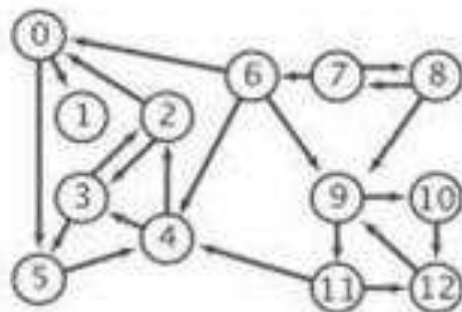
*A graph algorithm is an algorithm that takes more graph as input .*



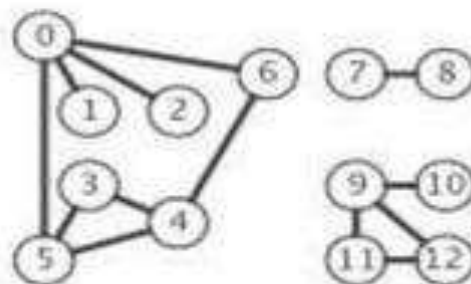


# Types of Graph algorithm

- ***Directed graph***



- ***Undirected graph***

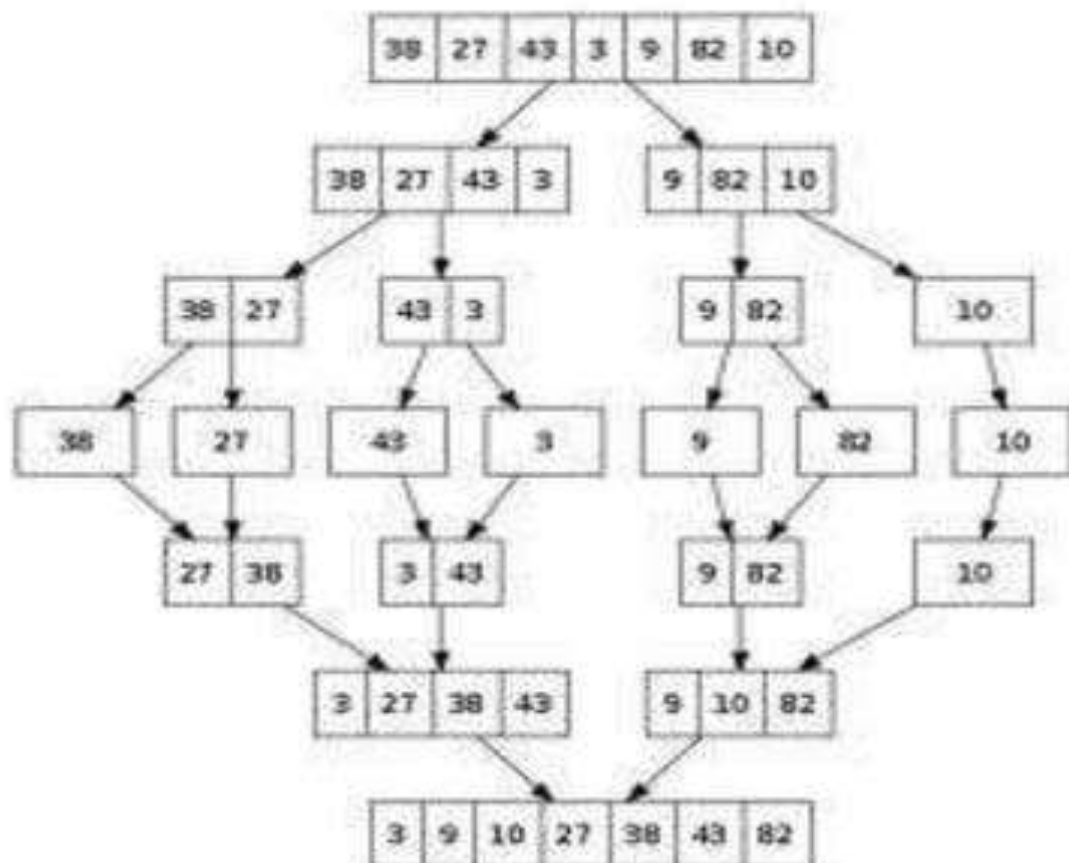


# Divide and Conquer

*The divide-and-conquer strategy solves a Problem by :*

- 1. Breaking it into subproblems that are themselves smaller instances of the same type of problem*
- 2. Recursively solving these*
- 3. Appropriately combining their answers*

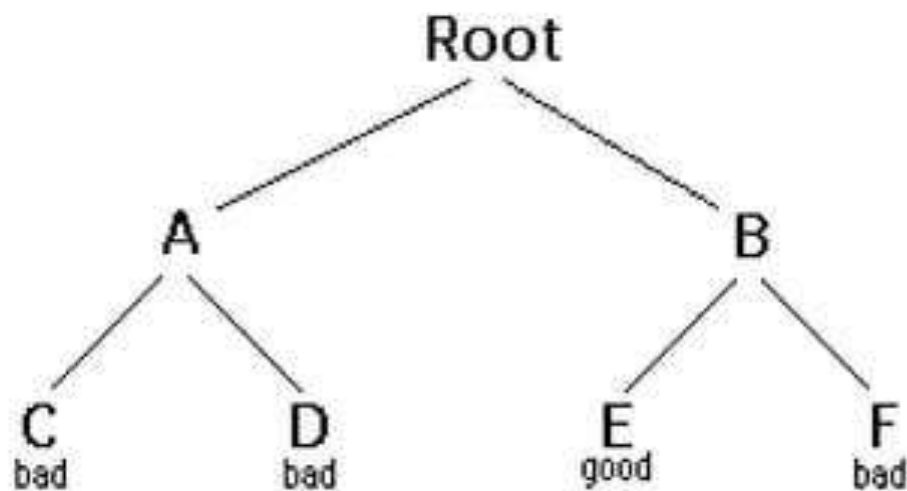
## Unsorted sequence



Sorted sequence

# Back Tracking

Backtracking is a general algorithm for finding all (or some) solutions to some computational problem.





# Greedy Algorithm

*1. What is greedy algorithm ?*

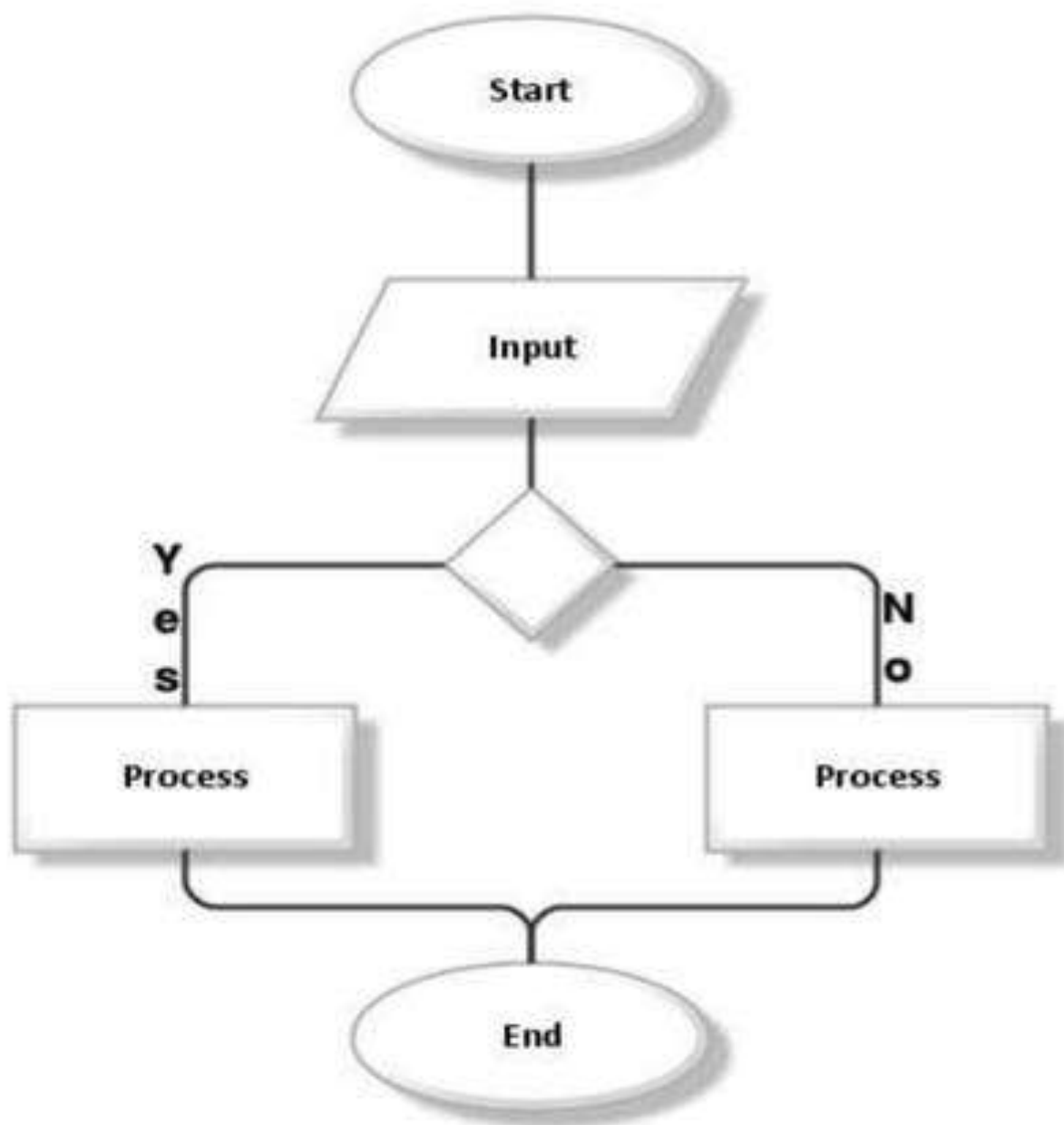
*2. Components of greedy algorithm*

- Candidate Set
- Selection Function
- Feasibility Function
- Objective Function
- Solution Function

*3. Drawback*

# Flowchart

*A **flowchart** is a type of diagram that represents algorithm , workflow or process, showing the steps as boxes of various kinds, and their order by connecting them with arrows.*





## Advantages of designing Algorithm

- ❖ *Ease of use*
- ❖ *Performance*
- ❖ *Scalability*
- ❖ *Stability*



*Thank You*

